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RESEARCH PAPER

Numerical simulation of molten pool behavior in wire arc additive manufacturing of overhanging structures using Inconel625

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Abstract

This paper investigates the quality control during the wire arc additive manufacturing (WAAM) process of nickel-based alloy structural components with overhang features. Firstly, a finite element model was developed in Fluent to study the temperature field evolution and melt pool flow of the nickel alloy components with overhang features in WAAM. A periodic coefficient φ was incorporated into the double-ellipsoidal heat source considering the periodic short-circuiting characteristics of CMT process. The effects of gravity, Marangoni force, electromagnetic force, and other forces on the flow behavior of the molten pool under different torch inclination angles are studied during WAAM of nickel alloy overhang structural components. With simulation results as references, WAAM experiments were performed on 60° inclined thin-walled overhang structural components. The results demonstrate that the established numerical model can effectively predict the forming quality of nickel-based alloy overhanging structures, providing a theoretical basis for improving forming accuracy and workpiece performance.

Keywords Overhanging structure · Wire arc additive manufacturing · Molten pool simulation · Nickel-based alloy · Forming quality control

1 Introduction

Nickel [1] and nickel-based alloys [2] possess excellent corrosion resistance, hot and cold workability, weldability, and low-temperature toughness, making them widely utilized in the chemical, petroleum, non-ferrous metal smelting, aerospace, and nuclear power structural manufacturing industries. Nevertheless, conventional manufacturing methods face challenges in efficiently producing workpieces with complex geometries. In contrast, additive manufacturing (AM) enables direct fabrication of intricately shaped components without reliance on complex tooling or multi-step assembly. Wire arc additive manufacturing (WAAM) [3] offers advantages including high deposition rates, efficient material utilization, relatively low equipment and production costs, the capacity to fabricate large-scale complex parts, and superior accessibility. Cold metal transfer (CMT) [4], a novel arc welding process, produces welds free of spatter and delivers high-quality deposits while maintaining low heat input, making it particularly suitable for forming thin-walled

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Highlights

- A multiphysics coupled model for Inconel625 overhanging structures was established, revealing the molten pool flow and heat transfer mechanisms under different inclination angles.
- The influence of the vortex structure in the molten pool under multi-field coupling on heat and mass transfer, as well as the uniformity of the solidification structure, was clarified.
- Support-free forming of a steep overhanging structure with a 60° inclination angle was successfully achieved.

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overhanging structures. Shi et al. [5] systematically optimized CMT-WAAM process parameters for 5087 aluminum alloy, achieving high-quality thin-walled components. Zhang et al. [6] successfully fabricated an Inconel625/80Ni1 composite structure via arc-based additive manufacturing, achieving a dense metallurgical interface and revealing composition gradients and performance evolution across the interface, with excellent bonding strength and fracture toughness. Tang et al. [7] optimized CMT-WAAM parameters for GH4169 alloy, producing predominantly columnar grains and attaining a maximum tensile strength of 642 MPa at a 45° inclination. Han [8] systematically addressed molten pool flow and forming accuracy in supportless overhanging structures through analysis of molten pool forces, collaborative optimization of process parameters and torch posture, and refined heat input control. Collectively, these studies demonstrate that CMT-WAAM holds broad application potential for fabricating high-performance metallic components and provides a robust theoretical basis for manufacturing complex structural parts.

Currently, there is significant demand for integrated fabrication of overhanging structures in industries such as aerospace. In wire arc additive manufacturing (WAAM), numerous studies have been conducted domestically and internationally to address the challenges of forming overhanging components. Kazanas et al. [9] utilized arc welding robots to perform all-position welding of thin-walled structures at arbitrary angles using CMT technology, adjusting the welding torch orientation to directly form complex overhanging geometries. Zargari and Laleh [10] successfully fabricated a 144-mm-tall thin-walled low-carbon steel component using a pulsed GMAW system with a rotating workpiece and stationary welding torch; by dynamically controlling the welding current and rotational speed, they produced components with qualified microstructure and mechanical properties. Yuan et al. [11] demonstrated the direct fabrication of complex overhanging structures without supports by intelligently optimizing the deposition orientation, overcoming the reliance of traditional WAAM on support structures. Li [12] proposed a WAAM process incorporating concrete supports for forming overhanging components, elucidating molten pool flow patterns and enabling the production of structurally sound overhangs with easily removable supports.

Analyzing the forming process of components through simulation of molten pool behavior is of significant importance. For instance, Fan et al. [13] developed a coupled arc-droplet-pool model for high-current MIG welding, demonstrating that the effects of metal vapor and an externally applied axial magnetic field on molten pool dynamics can enhance welding efficiency while maintaining weld quality. Nian et al. [14] established a three-dimensional numerical model accounting for free-surface deformation, revealing how ultrasonic vibration reduces penetration depth, accelerates steady-state forming, and refines grain size by regulating molten pool flow. Huang [15] proposed a method combining welding torch tilt control with staggered printing to address asymmetric melt pool morphology in arc additive manufacturing, providing guidance for high-quality component formation. Xu et al. [16] employed numerical simulations to reveal the evolution of melt pool flow from disorder to order during the SLM process, as well as thermal accumulation and remelting behavior during multi-track deposition. Huang et al. [17] combined real-time X-ray observation with VOF numerical simulations to establish a molten pool partition flow model, elucidating the influence of droplet transfer modes at varying wire feed rates on molten pool behavior in WAAM. Jin et al. [18] developed a finite element heat conduction model to accurately predict molten pool morphology in laser-arc hybrid welding of aluminum alloys. While current research predominantly focuses on carbon steel and aluminum alloys, which are prone to heat-affected cracking, studies on nickel-based alloys remain limited. In particular, the forming of overhanging structures lacks a robust theoretical foundation, and there is a shortage of simulation analyses addressing the temperature and flow fields within the molten pool of such components. Consequently, predicting molten pool behavior and ensuring forming precision and quality remain challenging.

This study integrates computer simulation with CMT wire arc additive manufacturing to investigate quality control in the fabrication of overhanging Inconel 625 structures. A finite element model was developed in Ansys-Fluent to simulate the evolution of the molten pool during the CMT additive manufacturing process. The research systematically examines process control strategies from the perspectives of heat-flow coupling and molten pool solidification, providing insights into the mechanisms governing overhang formation. The findings offer a theoretical foundation for enhancing the forming accuracy and mechanical performance of overhanging structural components in nickel-based alloy additive manufacturing.

2 Fundamentals of numerical simulation and model development

2.1 Basic assumptions

In this model, to balance simulation accuracy and computational efficiency, the following assumptions are made [19]:

1 Material exhibits isotropy.
2 Small-scale turbulent fluctuations within the melt pool are neglected, and the flow is assumed to be laminar.
3 Ignores the deposition effect of the filler metal.
4 By neglecting the complex variations in wire feed and arc behavior, the droplet formation process is simplified to the input of molten metal droplets at the velocity inlet. The arc's heating effect on the substrate is simplified to equivalent heating by a double-ellipsoidal heat source.

2.2 Control equations and molten pool forces

In the numerical simulation of welding thermal processes, the three fundamental conservation equations of fluid mechanics must be satisfied:

1) Mass conservation equation:

$$\frac{\partial \rho}{\partial t}+\frac{\partial(\rho u)}{\partial x}+\frac{\partial(\rho v)}{\partial y}+\frac{\partial(\rho w)}{\partial z}=0 \tag{1}$$

2) Momentum conservation equation:

$$\frac{\partial(\rho u)}{\partial t}+\frac{\partial(\rho u u)}{\partial x}+\frac{\partial(\rho u v)}{\partial y}+\frac{\partial(\rho u w)}{\partial z}=-\frac{\partial P}{\partial x}+\frac{\partial}{\partial x}\left(\mu\frac{\partial u}{\partial x}\right)+\frac{\partial}{\partial y}\left(\mu\frac{\partial u}{\partial y}\right)+\frac{\partial}{\partial z}\left(\mu\frac{\partial u}{\partial z}\right)+S_u \tag{2}$$

$$\frac{\partial(\rho v)}{\partial t}+\frac{\partial(\rho v u)}{\partial x}+\frac{\partial(\rho v v)}{\partial y}+\frac{\partial(\rho v w)}{\partial z}=-\frac{\partial P}{\partial y}+\frac{\partial}{\partial x}\left(\mu\frac{\partial v}{\partial x}\right)+\frac{\partial}{\partial y}\left(\mu\frac{\partial v}{\partial y}\right)+\frac{\partial}{\partial z}\left(\mu\frac{\partial v}{\partial z}\right)+S_v \tag{3}$$

$$\frac{\partial(\rho w)}{\partial t}+\frac{\partial(\rho w u)}{\partial x}+\frac{\partial(\rho w v)}{\partial y}+\frac{\partial(\rho w w)}{\partial z}=-\frac{\partial P}{\partial z}+\frac{\partial}{\partial x}\left(\mu\frac{\partial w}{\partial x}\right)+\frac{\partial}{\partial y}\left(\mu\frac{\partial w}{\partial y}\right)+\frac{\partial}{\partial z}\left(\mu\frac{\partial w}{\partial z}\right)+S_w \tag{4}$$

3) Energy conservation equation:

$$\frac{\partial(\rho H)}{\partial t}+\frac{\partial(\rho u H)}{\partial x}+\frac{\partial(\rho v H)}{\partial y}+\frac{\partial(\rho w H)}{\partial z}=\frac{\partial}{\partial x}\left(K\frac{\partial T}{\partial x}\right)+\frac{\partial}{\partial y}\left(K\frac{\partial T}{\partial y}\right)+\frac{\partial}{\partial z}\left(K\frac{\partial T}{\partial z}\right)+S_E \tag{5}$$

In the equation, u, v, and w represent the flow velocities of the liquid metal in the x, y, and z directions, respectively. ρ denotes the material density, P is the pressure, μ is the material viscosity, K is the thermal conductivity of the material, H is the enthalpy of mixing, S_u,S_v, and S_w are the source terms in the momentum equation, and S_E is the source term in the energy equation.

The fluid flow within the molten pool is primarily driven by a combination of forces [20], which are incorporated into the numerical model as source terms. A schematic illustrating these forces acting on the molten pool is presented in Fig. 1.

1. Surface tension

   The expression for surface tension is:

$$\gamma=\gamma_0+\frac{d\gamma}{dT}(T-T_m) \tag{6}$$

   In the equation, g_0 is the surface tension coefficient of the metal, dg/dT is the temperature gradient of surface tension, and T_m is the melting point temperature of the material.

2. Electromagnetic force

   The expression for the component of the electromagnetic force in the X direction is:

$$F_{magx}=-\frac{\mu_0 I^2}{4\pi^2\sigma_j^2 r}\exp\left(-\frac{r^2}{2\sigma_j^2}\right)\left[1-\exp\left(-\frac{r^2}{2\sigma_j^2}\right)\right]\left(1-\frac{z}{L}\right)^2\frac{x}{r} \tag{7}$$

   The expression for the component of the electromagnetic force in the Y direction is:

$$F_{magy}=-\frac{\mu_0 I^2}{4\pi^2\sigma_j^2 r}\exp\left(-\frac{r^2}{2\sigma_j^2}\right)\left[1-\exp\left(-\frac{r^2}{2\sigma_j^2}\right)\right]\left(1-\frac{z}{L}\right)^2\frac{y}{r} \tag{8}$$

   The expression for the component of the electromagnetic force in the Z direction is:

$$F_{magz}=\frac{\mu_0 I^2}{4\pi^2 L r^2}\left[1-\exp\left(-\frac{r^2}{2\sigma_j^2}\right)\right]^2\left(1-\frac{z}{L}\right) \tag{9}$$

   In the equation: μ_0 is the magnetic permeability of free space, I is the welding current, σ_j is the arc current density distribution parameter, r is the arc distribution radius, $r=\sqrt{(x-ut)^2+y^2}$, and L is the substrate thickness.

3. Thermal buoyancy

   This model approximates the effect of buoyancy using the Boussinesq approximation. The expression for buoyancy is calculated as follows:

$$F_f=-\rho g\beta_0 \triangle T \tag{10}$$

   In the equation, p is density, g is gravitational acceleration, β_0 is the thermal expansion coefficient of the metal, and ΔT is the temperature difference.

4. Gravity

   The expression for gravity is:

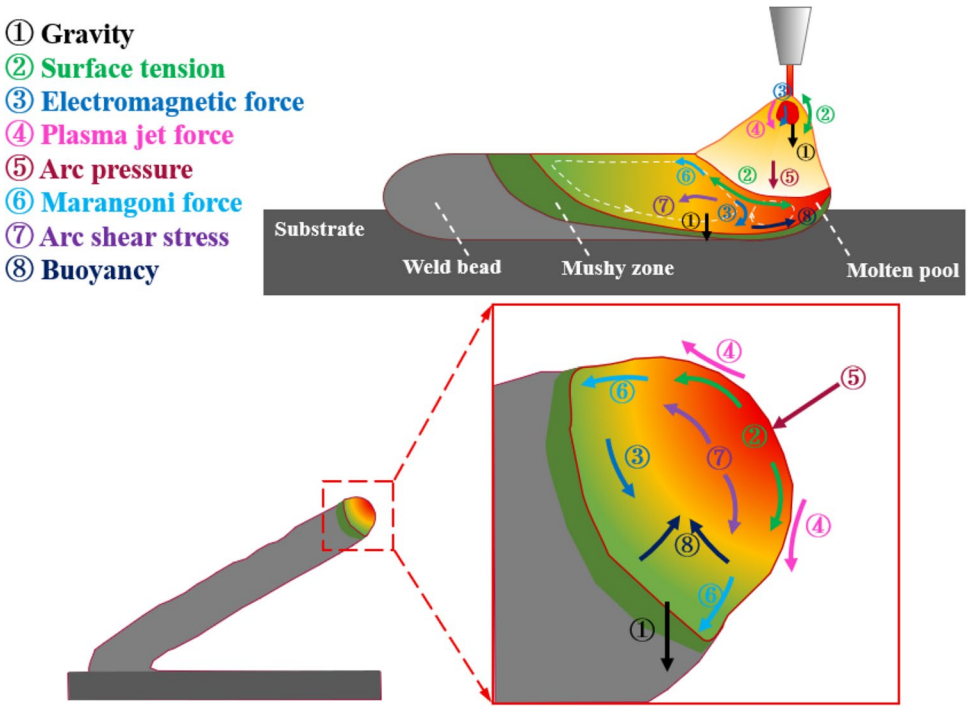


Fig. 1 Force diagram of the molten pool

$$G = mg \tag{11}$$

In the equation, $g = 9.8\ \mathrm{m/s^2}$.

2.3 Thermal source model

The physical model of the welding heat source encompasses two key aspects: first, the fraction of thermal energy from the heat source that is transferred to the workpiece, and second, the spatial distribution of this transferred heat across the workpiece surface. Establishing an accurate physical model of the welding heat source is therefore a prerequisite for reliable numerical simulations.

Owing to the short-circuiting characteristics of CMT welding, the arc heat input during the process is relatively low. The arc energy is concentrated and moves along the welding path, exhibiting an uneven distribution between the front and rear regions, which results in an asymmetrical heating zone. Consequently, a double-ellipsoidal heat source [21] is selected to more accurately simulate the effect of the arc. In practice, the CMT welding cycle is determined by the welding current; for instance, at a current of 129 A, the measured CMT cycle is 18 ms. To account for the periodic nature of the CMT process, a periodic coefficient φ is incorporated into the double-ellipsoidal heat source model, as illustrated in Fig. 2, providing a more realistic representation of arc heat input during the welding process.

The periodic function φ can be represented as:

$$\varphi = \frac{1 + sin[369(t + 0.006)]}{2} \tag{12}$$

The formulas for the front and rear halves of a double-ellipsoid are as follows:

$$q_f = \frac{6\sqrt{3}\varphi f_f Q}{\pi^{\frac{3}{2}} a_f b_h c_h} \exp\left[-3\left(\frac{x^2}{a_f^2} + \frac{y^2}{{b_h}^2} + \frac{z^2}{{c_h}^2}\right)\right] \tag{13}$$

$$q_r = \frac{6\sqrt{3}\varphi f_r Q}{\pi^{\frac{3}{2}} a_r b_h c_h} \exp\left[-3\left(\frac{x^2}{a_r^2} + \frac{y^2}{{b_h}^2} + \frac{z^2}{{c_h}^2}\right)\right] \tag{14}$$

In the formula, α_f, α_r, b_h, and C_h represent the semi-axes of the dual ellipsoids; F_f and F_r denote the respective shares of heat input in the front and rear ellipsoids, where $f_f + f_r = 2$; q_f and q_r are the energy distribution coefficients for the front and rear sections of the dual ellipsoid heat source.

2.4 Boundary conditions

The overhanging model has dimensions of 30 mm × 30 mm × 18.5 mm. The computational domain is illustrated in Fig. 3, and the boundary conditions for each face are summarized in Table 1. In the model, the molten droplet parameters are specified at the velocity inlet boundaries, while wall boundaries define the heat dissipation conditions within the domain. A regular hexahedral mesh was adopted for spatial discretization, with mesh refinement optimized at the welded joints. The refined hexahedral elements have an edge length of approximately 0.2 mm, while the

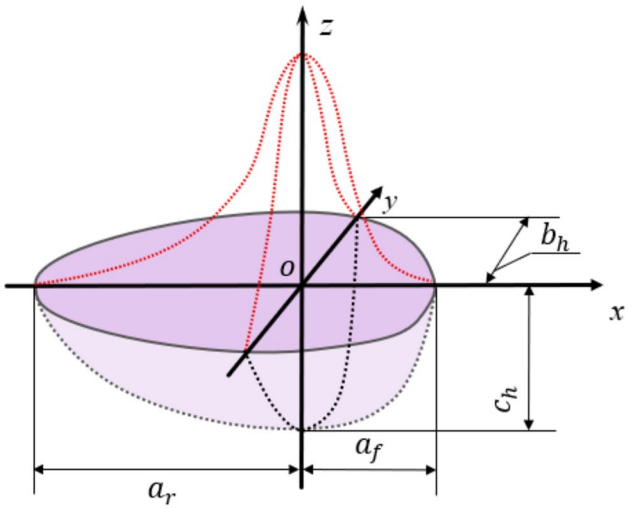


Fig. 2 Double-ellipsoidal heat source model

remaining edges are coarsened to roughly 0.5 mm, resulting in a total of 285,600 discrete elements. The pressure-based solver was employed for the simulations, using the SIMPLE algorithm [22] for numerical processing. Discretization was performed with a second-order upwind scheme, and the initial time step for the computations was set to 1×10^{-4} s.

2.5 Thermophysical properties of materials

This study selected the nickel-based alloy Inconel625 as the base material for arc welding of overhanging structural components. The thermophysical parameters of Inconel625 [23] are shown in Table 2.

The density ρ, specific heat capacity C_p, and thermal conductivity λ of the material vary with temperature [24], as shown in Fig. 4.

Table 2 Thermophysical properties of Inconel625

Physical quantity	Symbol	Unit	Value
Ambient temperature	T_0	K	300
Solidus temperature	T_s	K	1290
Liquidus temperature	T_l	K	1350
Latent heat of fusion	H_m	$J \cdot kg^{-1}$	2.7×10^5
Coefficient of thermal expansion	β	K^{-1}	14.1×10^{-6}
External emissivity	ε	/	0.4
Coefficient of heat convection	h	$W \cdot m^{-2} \cdot K^{-1}$	100
Coefficient of surface tension	γ	$N \cdot m^{-1}$	1.65
Viscosity	μ	$kg \cdot (m \cdot s^{-1})^{-1}$	0.003

3 Simulation results and analysis

3.1 Experimental validation of single-layer single-channel numerical simulation results

To ensure the accuracy of the simulation results, single-layer, single-pass welding experiments were conducted using a CMT welder. Precise control of heat input and welding current was maintained: sufficient temperature was required to achieve complete penetration of the filler wire, while excessive heat was avoided to prevent undesired molten pool flow. By systematically optimizing the wire feed speed and welding speed, the parameter combination yielding optimal weld bead formation was selected, as summarized in Table 3. The obtained single-layer, single-track specimen melt pool morphology profile is compared with the simulation results, as shown in Fig. 5. In the experiments, the measured weld width, depth, and height were 5.6 mm, 0.9 mm, and 2 mm, respectively, whereas the corresponding simulated values were 5.62 mm, 0.93 mm, and 1.98 mm, resulting in error rates of 0.36%, 3.33%, and 1.00%, respectively. The close agreement between experimental and simulated molten pool

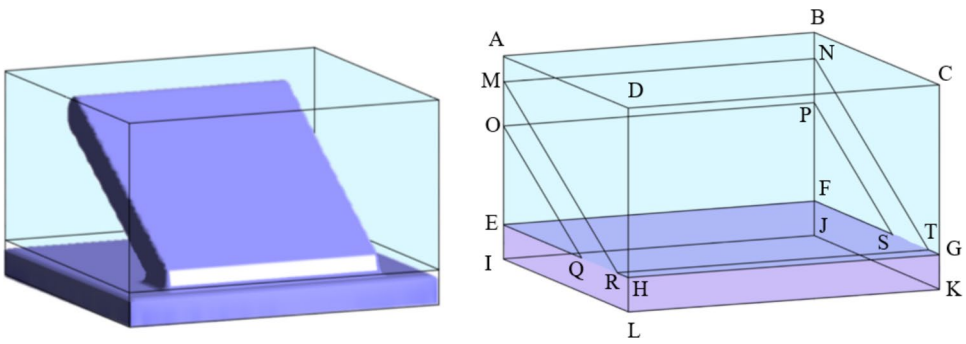


Fig. 3 Numerical simulation computational domain

Table 1 Boundary types of the computational model

Type	INLET	OUTLET	WALL
Boundary	ABCD	DCGH, MRQO, NTSP, ABFE	EFJI, FGKJ, HGKL, EHLI, IJKL

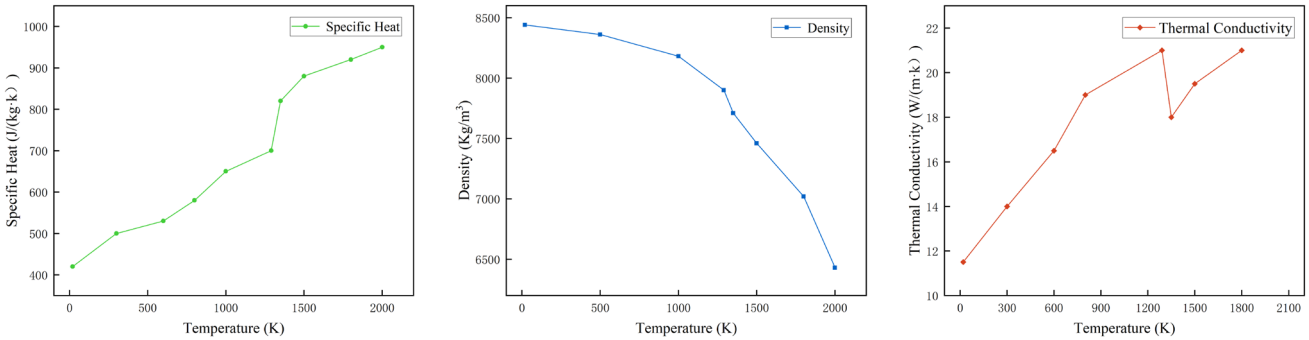


Fig. 4 Curve of thermal physical parameter variation

Table 3 Parameters for the selected group

Wire feed speed (mm/s)	Welding speed (mm/s)	Electric current (A)	Voltage (V)
3.7	7	93	14.1

profiles demonstrates the validity of the chosen parameter set for numerical simulation of the temperature field in Inconel625 WAAM, providing reliable initial conditions for subsequent simulations and experiments on complex overhanging structures.

3.2 Evolution of molten pool morphology

Figure 6 illustrates the topography and temperature field distribution of a single-layer, single-track molten pool at various time points. At the initial stage ($t = 0.005$ s), the high-temperature arc transfers energy to the substrate surface via radiation, causing the substrate temperature to begin rising. Between $t = 0.01$ s and 0.3 s, the molten pool enters a dynamic development stage: the temperature gradient broadens, and arc pressure induces a noticeable depression at the center of the pool. During this stage, multiphysics coupling effects dominate the behavior of the molten pool. The temperature gradient at the surface drives Marangoni convection, while Lorentz forces generated by the welding current induce eddy currents within the molten pool. The influence of gravity also becomes increasingly significant. At a later stage ($t = 0.9$ s), the molten pool reaches a quasi-steady state in both temperature and geometry. Heat input and substrate heat dissipation approach equilibrium, and the liquid metal progressively solidifies into the weld structure. Simultaneously, internal flow stabilizes as temperature homogenization progresses, ultimately forming a weld cross-section with a characteristic depth-to-width ratio.

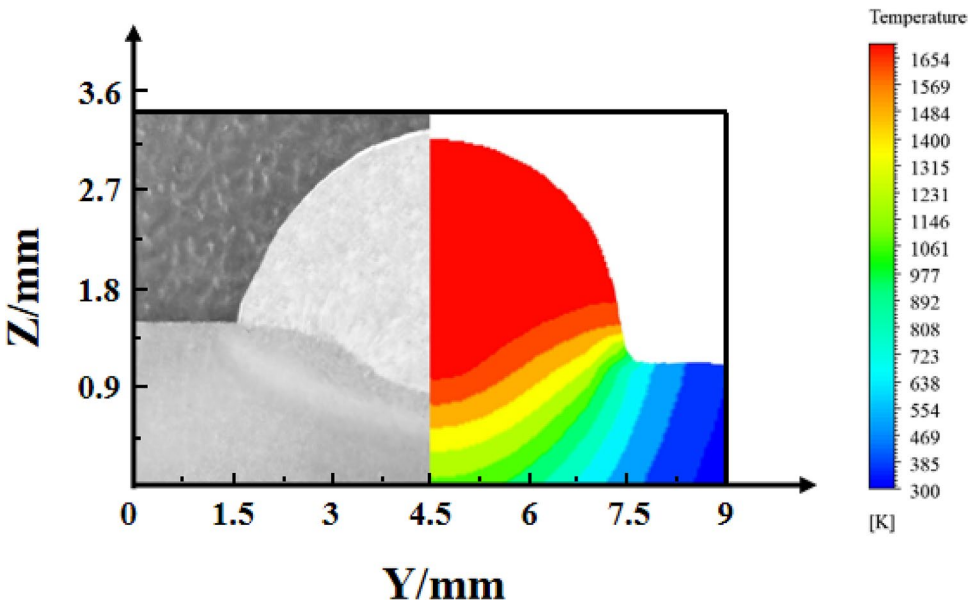


Fig. 5 Comparison of weld fusion lines

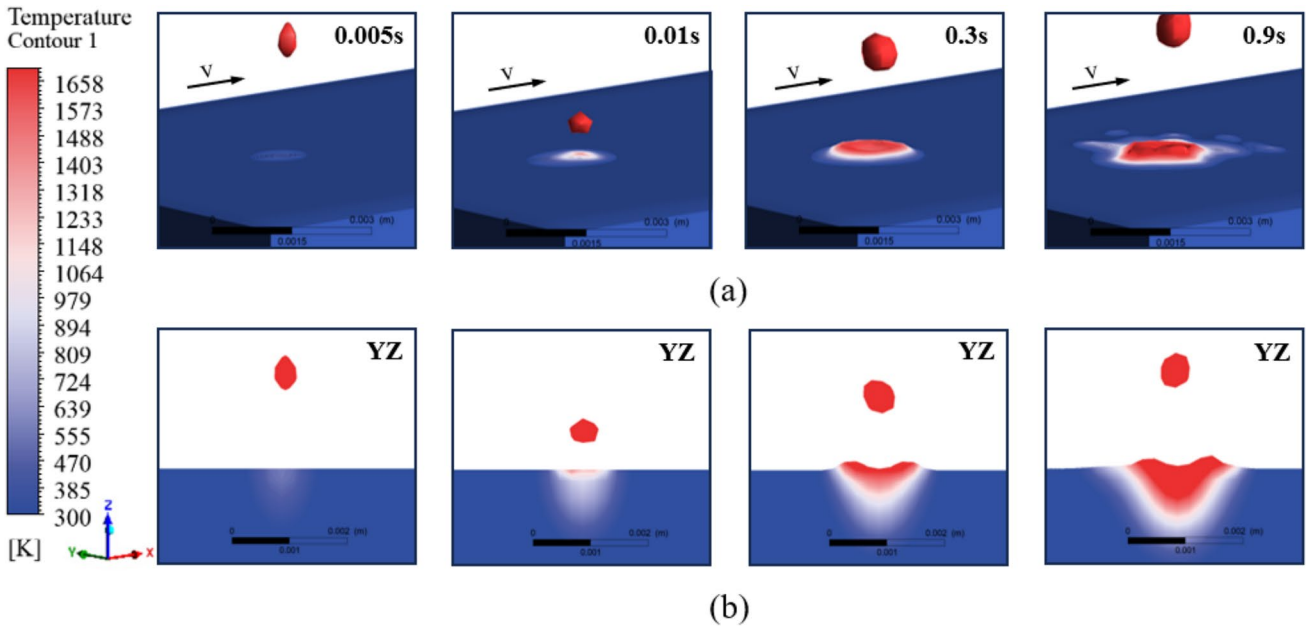


Fig. 6 Melt pool temperature field contour maps: **a** overall contour map, **b** YZ cross-section contour map

3.3 Simulation results of overhang additive manufacturing at different torch angles

In the numerical simulation of additive manufacturing for cantilever structures, each weld bead was designed with a length of 30 mm. The simulation time step was dynamically adjusted according to the deposition rate, with the deposition position spanning from 0 to 30 mm along the x-axis. During the simulation, molten droplets were injected at inclination angles of 30° and 60° relative to the velocity inlet, depositing onto the surface of a preformed thin-walled substrate and undergoing repeated welding cycles. As additional layers were sequentially deposited, the target cantilever structure gradually took shape.

3.3.1 Analysis of the forming process for a 30°-angled straight wall

Figure 7 presents the numerical simulation results of the final-layer welding process for the 30° inclined straight-wall structure, showing both the macroscopic morphology of the molten pool and the temperature distribution across characteristic cross-sections at different time points. The analyzed cross-section is parallel to the inclination direction of the straight wall and is located at the center of the cantilever structure, providing a representative view of the thermal and geometric evolution during the final deposition stage.

During the initial stage ($t = 0.005$–0.5 s), as molten droplets impact the substrate, the molten pool exhibits a small, localized volume. The cross-sectional temperature field is highly inhomogeneous, with a localized high-temperature region at the center and rapid cooling toward the periphery. The central region, where the molten droplet contacts the workpiece, exceeds the melting point of the metal, while temperatures decrease sharply at the edges. This behavior arises from concentrated heat input at this stage, with limited thermal diffusion, preventing the establishment of stable heat transfer pathways. Upon entering the steady-state phase ($t = 0.65$–1.2 s), the molten pool expands along the welding direction, and the cross-sectional temperature field gradually becomes more uniform. The central high-temperature zone elongates along the straight-wall extension, while the interface between the low-temperature solidified region and the high-temperature molten zone remains distinct and continuous. This indicates that once the heat input rate reaches equilibrium with heat diffusion and radiation, the heat transfer pathways stabilize, resulting in a relatively uniform temperature field across the welded component.

Figure 8 illustrates the flow field distribution on the molten pool surface and within a characteristic cross-section at different time points. At the initial moment ($t = 0.005$ s), concentrated arc heating rapidly forms the molten pool, and Marangoni forces drive the surface melt from the high-temperature region toward the cooler edges. During this stage, the flow field is dominated by outward expansion along the surface layer. At $t = 0.25$ s, the molten pool enlarges, and the coupled effects of multiple forces govern the flow field evolution. The gravitational component along the slope drives molten material toward the downhill side, elongating the molten pool and increasing flow velocity in

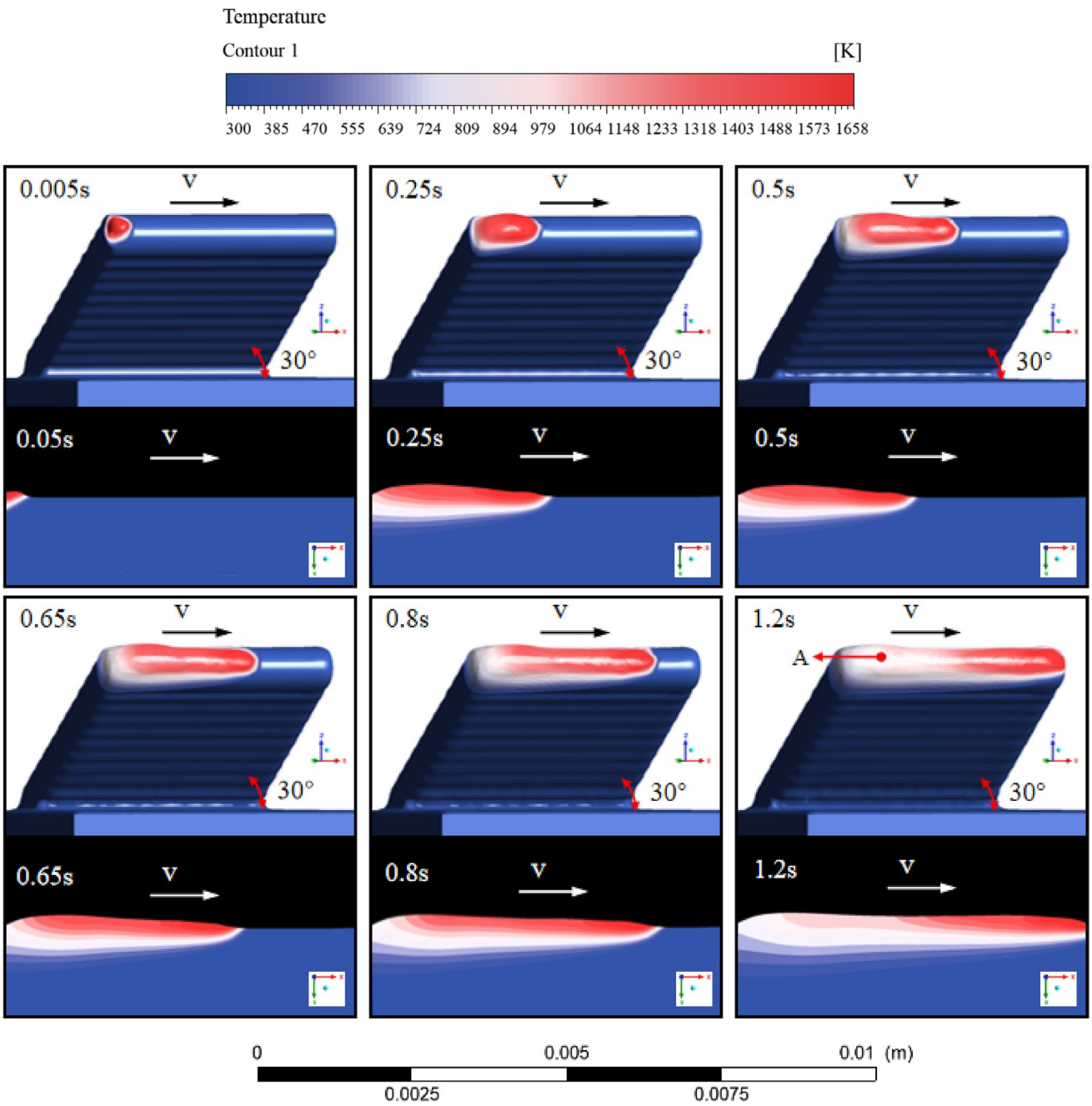


Fig. 7 Molten pool morphology and temperature distribution of the 30°-angled thin-walled structure at different time instants

that region, while the uphill side exhibits backflow due to the opposing gravitational component. The vertical gravitational component induces buoyant convection, causing high-temperature melt to rise and cooler melt to sink, forming internal circulation. On the uphill side, the return flow interacts with the mainstream on the downhill side, generating a structured vortex with a distinct rotational direction due to velocity gradient and momentum imbalances. This vortex reflects a dynamic equilibrium of multiple forces. By $t = 0.5$ s, the quasi-steady state is reached, with heat input and dissipation in dynamic balance, resulting in a stable flow field and a more compact vortex structure. This analysis elucidates the physical mechanisms governing the evolution of the flow field—from initial Marangoni-driven surface motion to multi-force-induced asymmetric vortex formation—providing critical insights for flow regulation and process optimization in the additive manufacturing of overhanging structures.

The temperature and flow velocity profiles at point A in Fig. 7 are presented in Fig. 9.

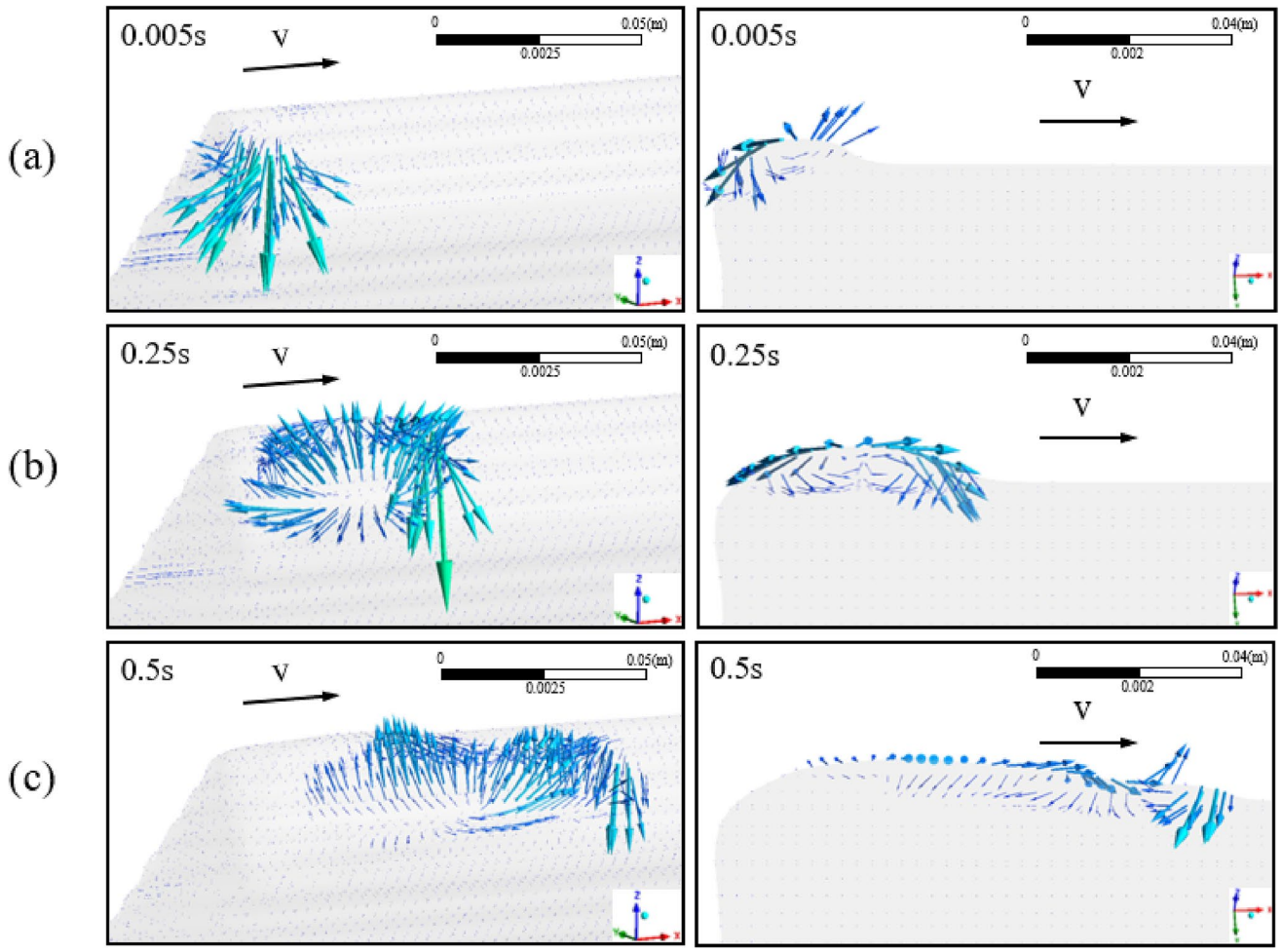


Fig. 8 Flow field distribution of the molten pool surface and characteristic cross-sections of the 30°-angled straight wall at different time instants (**a**) $t = 0.005$s; (**b**) $t = 0.25$s; (**c**) $t = 0.5$s

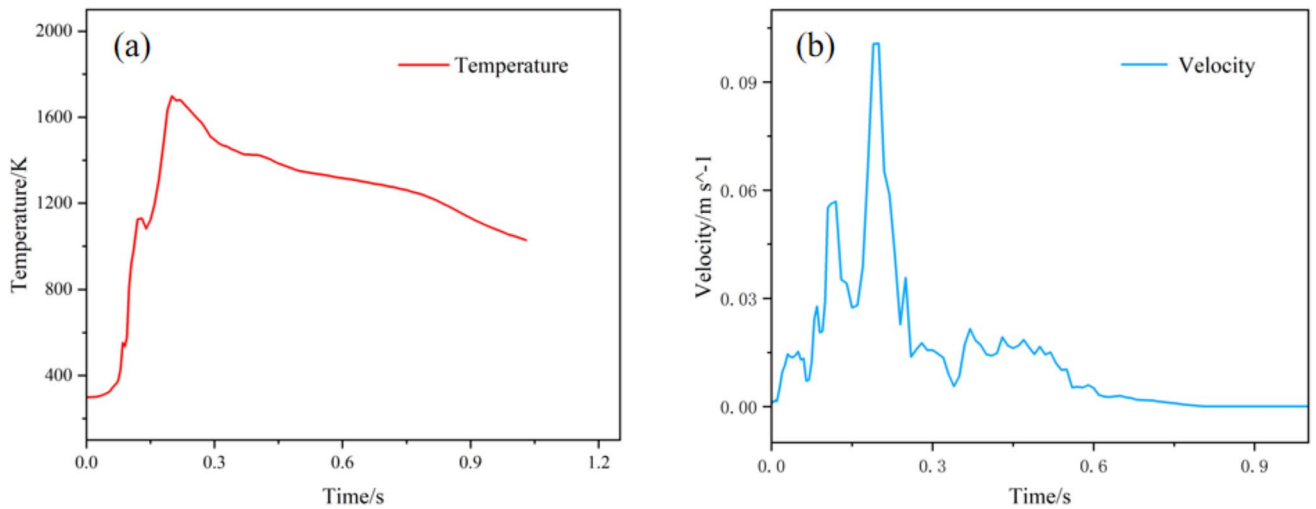


Fig. 9 Variations of temperature (**a**) and flow velocity (**b**) with time at point A

The temperature curve indicates that, during the initial stage of welding, the temperature at the measurement point rises rapidly to a peak of 1658 K. This localized high temperature results from concentrated heat input, which limits its heat diffusion and is characteristic of the early molten pool stage. As the molten pool expands and heat dissipation increases, the temperature at point A decreases gradually, and the temperature field becomes more uniform. During the steady-state stage, heat input and dissipation achieve dynamic equilibrium, leading to only minor temperature fluctuations. These observations confirm the evolution of the temperature field in the inclined straight-arm welding process from an initial non-uniform distribution to a quasi-steady state.

The flow velocity profile at point A shows that, during the initial stage of welding, the molten pool exhibits rapid motion primarily driven by surface tension gradients, with the velocity quickly reaching 0.06 m/s, dominated by the Marangoni effect. As the molten pool develops, gravitational components due to inclination and buoyancy-driven

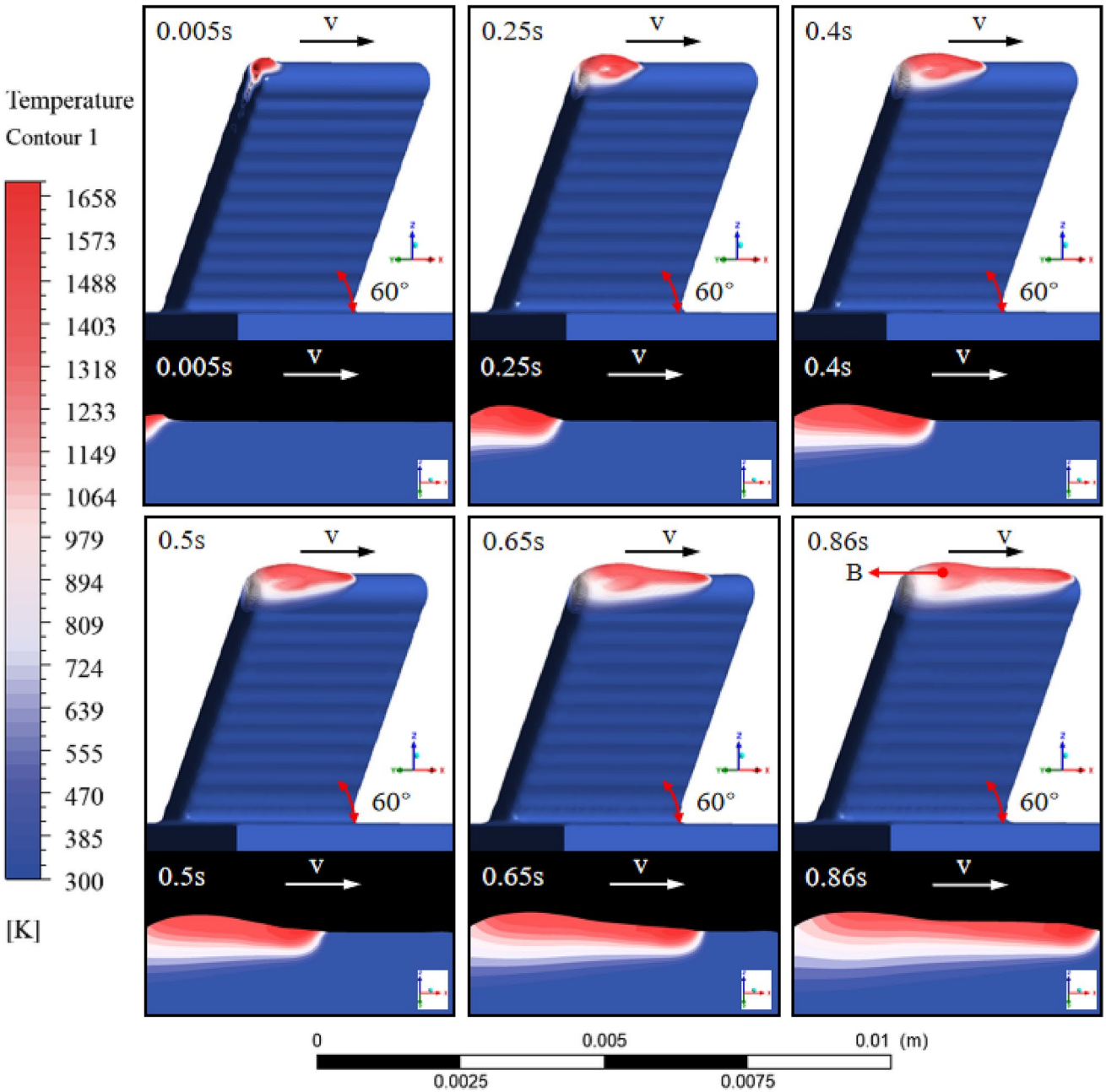


Fig. 10 Molten pool morphology and temperature distribution of the 60°-angled thin-walled structure at different time instants

convection jointly influence molten metal motion. Interactions within the flow field result in a peak velocity of approximately 0.10 m/s and the gradual formation of an asymmetric vortex pattern. Upon entering the quasi-steady stage, the molten pool flow intensity decreases markedly, and the velocity stabilizes, illustrating the evolution of molten pool flow from a single-force-dominated regime to a multi-force-coupled steady-state behavior.

3.3.2 Analysis of the forming process for a 60°-angled straight wall

In the numerical simulation of a 60° inclined straight-wall overhang structure, the evolution of the molten pool temperature field is shown in Fig. 10. During the initial stage ($t = 0.005$–0.25 s), after deposition of molten droplets onto the substrate, a small-volume molten pool forms. The cross-sectional temperature distribution is highly non-uniform, primarily influenced by directional heat conduction associated with the large 60° inclination. At this stage, heat input is highly concentrated, and heat transfer pathways have not yet stabilized, resulting in pronounced and irregular temperature gradient fluctuations. During the intermediate stage ($t = 0.4$–0.5 s), the molten pool expands, and temperature inhomogeneity becomes more evident. The high-temperature zone elongates along the straight-wall extension, with the central region exceeding the melting point, while the boundary between high- and low-temperature regions sharpens and the peripheral temperature gradient becomes smoother. In the quasi-steady phase ($t = 0.65$–0.86 s), the temperature field stabilizes while maintaining a distinctly asymmetric morphology. The central high-temperature zone exhibits regularized contours, with a stable radial temperature decay from center to periphery, indicating equilibrium between heat input and dissipation. The interface between the low-temperature solidified zone and the molten region remains continuous and well-defined. Heat conduction is primarily oriented along the wall direction, ultimately

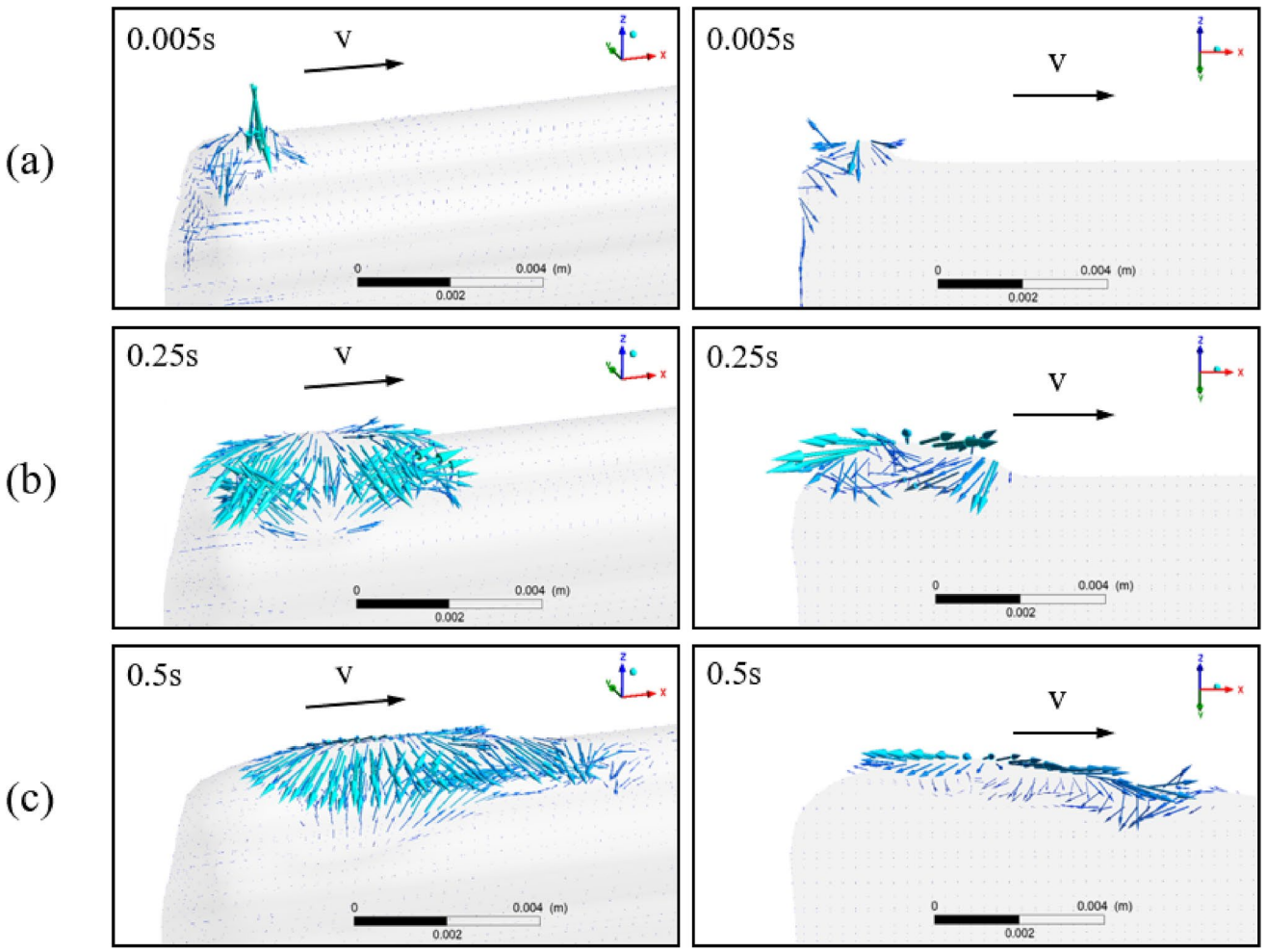


Fig. 11 Flow field distribution of the molten pool surface and characteristic cross-sections of the 60°-angled straight wall at different time instants (**a**) $t = 0.005$s; (**b**) $t = 0.25$s; (**c**) $t = 0.5$s

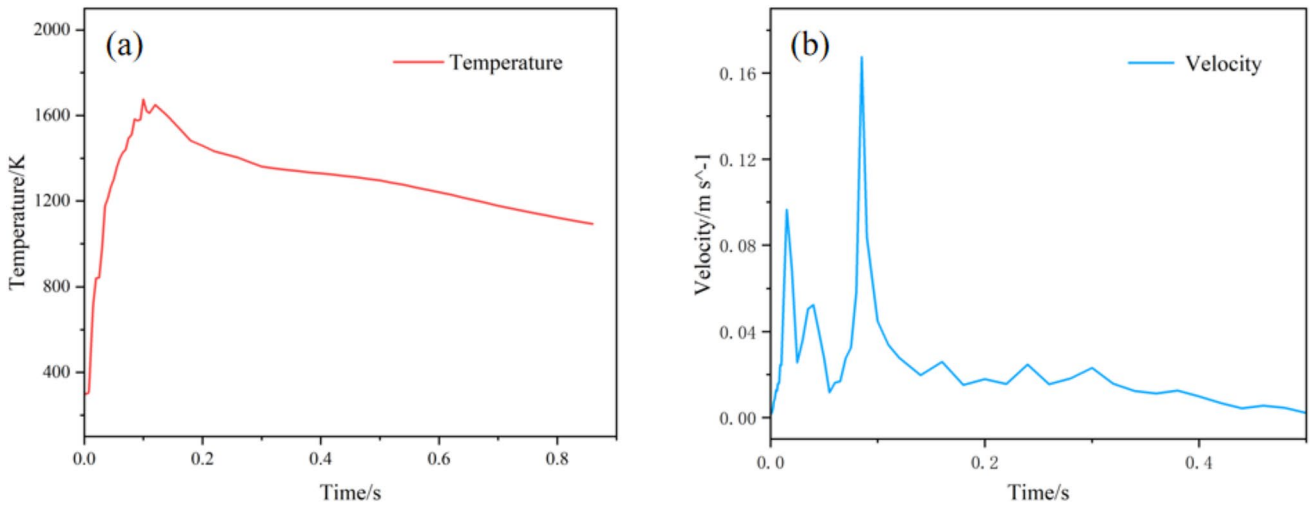


Fig. 12 Variations of temperature (**a**) and flow velocity (**b**) with time at point B

Table 4 Welding wire chemical composition (*wt%*)

Element	Ni	Cr	Mo	Nb + Ta	Fe	Co	C
Content	58	21	9	3.25	5	1	0.1

establishing a stable temperature field distribution adapted to the 60° inclined configuration.

Figure 11 illustrates the flow velocity distribution at the free surface and within a characteristic cross-section of the molten pool at various time points. At the initial moment ($t = 0.005$ s), the flow is dominated by outward surface expansion driven primarily by Marangoni forces, with gravitational effects not yet fully developed. During the transition phase ($t = 0.25$ s), the molten pool expands significantly, and the gravitational component along the slope becomes the dominant force, driving downward flow toward the slope side and producing a distinctly asymmetric flow field. Surface Marangoni flow converges downslope under gravitational modulation, while arc pressure drives outward motion, and electromagnetic forces begin to induce contraction and rotation within the melt. Multi-force coupling at this stage generates gravitational counterflow and velocity shear on the uphill surface, initiating the formation of a recirculation zone. In the quasi-steady stage ($t = 0.5$ s), momentum exchange and velocity discontinuities in the lower-middle region between the downslope mainstream and counterflow or wall-constrained flow promote streamline bending and rotation. This flow separation and shear instability lead to vortex formation around an instantaneous center, which serves as the core medium for heat and mass transfer within the molten pool, directly affecting solidification microstructure uniformity and defect distribution. At a 60° inclination, the slope-aligned gravitational component produces pronounced downhill flow, with downhill velocities approximately twice those of the uphill side. Despite this, the low heat input and rapid solidification characteristics of the CMT process, combined with surface tension and electromagnetic constraints, prevent molten pool collapse and maintain stable weld formation. Overall, this process reveals the evolution of molten pool flow from Marangoni-dominated motion to gravity-dominated asymmetric vortices, ultimately achieving stable multi-force-coupled vortex structures, reflecting

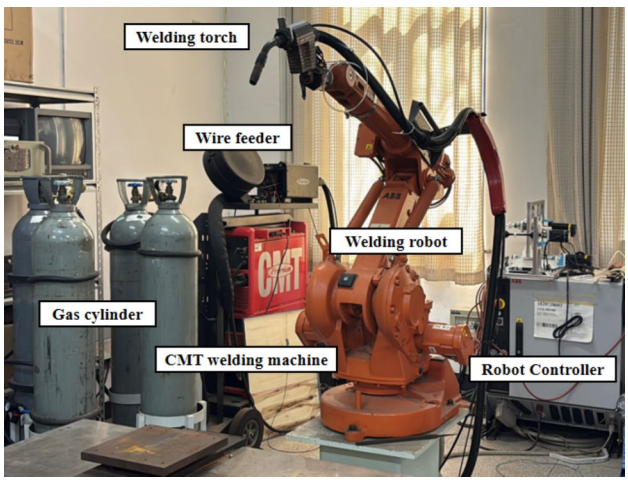


Fig. 13 Experimental system

Table 5 Substrate chemical composition (*wt%*)

Element	C	Mn	Si	S	P
Content	0.22	0.65	0.3	0.05	0.045

the complex flow behavior in high-angle overhang additive manufacturing.

Characteristic point B in Fig. 10 was selected to extract temperature and flow velocity responses throughout the welding process, as presented in Fig. 12.

The temperature profile at point B indicates that, during the initial stage of welding, the temperature rises rapidly to a peak of 1660 K, corresponding to the non-steady-state stage characterized by concentrated heat input and limited heat diffusion. This behavior aligns with the localized high-temperature and pronounced temperature gradient fluctuations observed in the early molten pool. As the pool expands, the temperature gradually decreases, and the high-temperature zone elongates along the extension direction of the straight arm, reflecting the gradual establishment of dynamic equilibrium between heat input and heat dissipation. During the quasi-steady stage, temperature variations become smoother, confirming the evolution of the temperature field in the 60° inclined straight arm from an initial non-uniform distribution to a stable asymmetric configuration.

The flow velocity profile at point B shows that, during the initial stage of welding, the velocity rapidly rises to 0.10 m/s, corresponding to outward surface flow dominated by the Marangoni effect. The velocity then peaks at 0.17 m/s at $t = 0.15$ s, as the gravitational component along the slope strengthens and interacts with Marangoni and electromagnetic forces, producing an asymmetric flow field biased toward the downhill side. Velocity shear between the uphill backflow and the main downslope flow leads to the maximum flow intensity, with the downhill velocity approximately twice that of the uphill side, consistent with the asymmetric features observed in the flow vector field. Upon reaching the quasi-steady stage, the flow velocity decreases and gradually stabilizes, reflecting the establishment of stable vortical circulation. This evolution demonstrates that the molten pool flow transitions from intense asymmetric motion to stable vortex-dominated circulation, further confirming that the low heat input characteristic of the CMT process effectively suppresses melt pool collapse.

The inclination angle alters the gravitational component along the slope, thereby affecting the heat transfer characteristics and force conditions of the molten pool, which leads to different evolution behaviors of temperature and flow velocity under the 30° and 60° working conditions. At a 60° inclination, the gravitational component along the slope is relatively large. Heat conduction exhibits strong directionality with significant heat accumulation, resulting in a higher temperature peak and a more pronounced asymmetric temperature field distribution. Under these force conditions, the downward flow of molten metal is enhanced, and the multi-force coupling effect is intensified, leading to higher flow velocity and stronger flow fluctuations. At a 30° inclination, the gravitational effect is relatively weak. Heat dissipates more uniformly, and temperature variations are comparatively moderate. The molten flow is mainly governed by surface tension, resulting in a lower overall flow velocity and a more stable flow pattern.

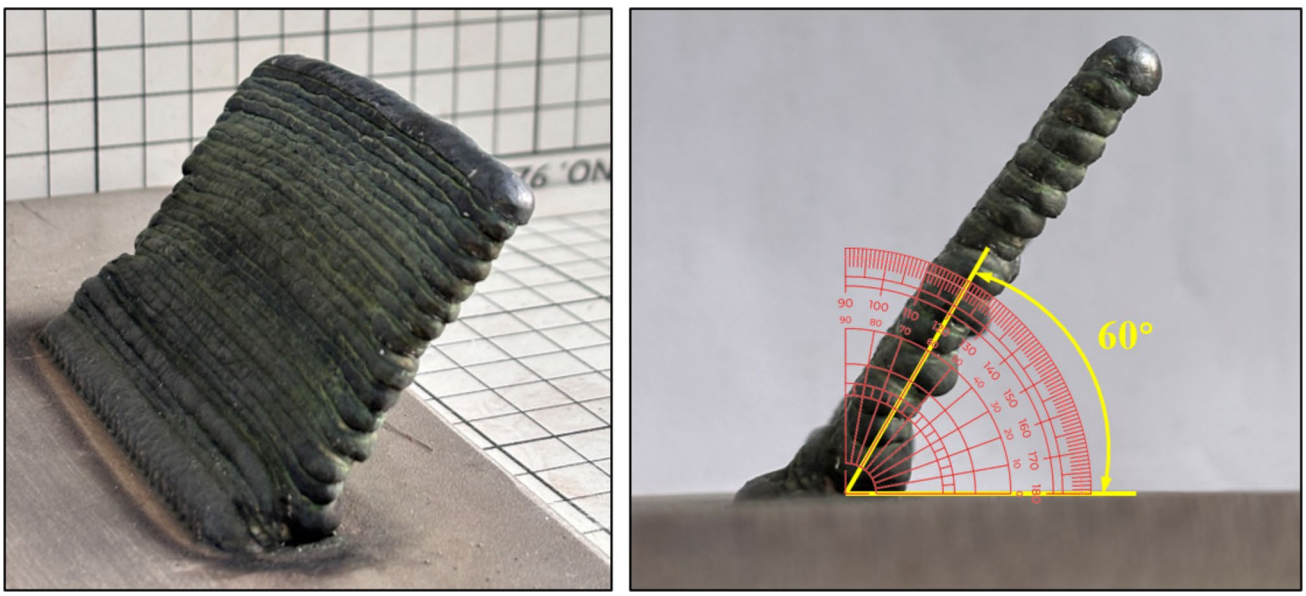


Fig. 14 Specimen with a straight wall overhang structure inclined at 60°

4 Inconel625 overhanging structure additive manufacturing experiment

4.1 Experimental system

The experiments were conducted using a Fronius TPS4000 welding system in conjunction with an ABB IRB 1410 six-axis industrial robot, with the robot serving as the motion actuator. The integrated experimental setup is illustrated in Fig. 13. The Inconel625 welding wire used in the experiment has a diameter of 1.2 mm and is designated as THQ-50C. The chemical composition of the welding wire is shown in Table 4. The substrate material was Q235 low-carbon steel, with its chemical composition listed in Table 5. High-purity argon (99.9%) was used as the shielding gas, with a controlled flow rate of 15–30 L/min to ensure stable welding conditions.

4.2 Experimental results

The test specimen obtained from the wire arc additive manufacturing of a 60° inclined straight-wall overhanging structure, fabricated using welding methods and process parameters consistent with the simulations, is shown in Fig. 14. Macroscopic analysis of the specimen indicates excellent forming quality, with complete structures and well-defined contours, suggesting effective control of molten pool flow and thermal fields during the process. The downslope side exhibits a uniformly spread thin-layer structure, while the upslope side shows well-formed edges and strong interlayer bonding. These characteristics are consistent with the numerical simulation predictions, including gravity-driven downward displacement of the molten pool and the formation of stable vortex flow under multi-force coupling. The well-formed specimen demonstrates that the established numerical model can accurately characterize the multiphysics coupling behavior and forming mechanisms of high-inclination overhang structures in wire arc additive manufacturing, thereby validating the simulation approach and providing a reliable theoretical basis for process optimization and forming control.

5 Conclusion

This study addresses the forming challenges of overhanging structures in CMT wire arc additive manufacturing of Inconel625 nickel-based alloy. A thermo-fluid-mechanical numerical model incorporating multiphysics coupling was developed to investigate the dynamic behavior of the molten pool and the effects of process parameters during the forming process. The accuracy of the model was validated through experimental results. The main conclusions are summarized as follows:

1) A numerical simulation model for CMT wire arc additive manufacturing of overhanging structures using Inconel 625 nickel-based alloy was developed. Validation through single-layer, single-pass experiments demonstrated good agreement between simulated and experimental molten pool morphologies, confirming the accuracy and reliability of the model.
2) The study elucidates the morphological evolution of the molten pool during the additive manufacturing of overhanging structures. In the initial stage, the molten pool exhibits a localized distribution with pronounced temperature gradients. During the transition stage, enhanced multiphysics coupling drives molten flow through the combined effects of gravity, Marangoni forces, and electromagnetic forces. In the quasi-steady stage, heat input and dissipation reach equilibrium, resulting in a stable molten pool morphology.
3) The influence of welding torch tilt angle on molten pool behavior was clarified. At a 30° tilt angle, Marangoni forces dominate the surface flow, while gravitational effects are relatively weak, resulting in a symmetric vortex structure. In contrast, at a 60° tilt angle, the gravity-induced slope component becomes the primary driving force, causing molten pool displacement and the formation of an asymmetric vortex. This asymmetric vortex serves as the main medium for heat and mass transfer, directly affecting the uniformity of the solidified microstructure.
4) A 60° inclined overhanging structure was successfully fabricated, exhibiting excellent formability, well-defined contours, and no macroscopic defects. This confirms that the numerical model can accurately characterize the multiphysics coupling behavior during the forming process of high-angle overhanging structures.

Author contribution All authors contributed to the study conception and design. Junjie He: writing—review and editing, investigation, data curation. Ruiting Qu: validation, methodology, writing—original draft. Tianqi Wang: writing—review and editing, investigation. Lifeng Du: data curation. Yifan Liang: data curation. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Declarations

Competing interests The authors declare no competing interests.

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